

Stats Library Documentation

I created a StatsLib class that provides methods for computing mean, median, modes, standard deviation, probability distributions, and methods for determining independence and dependence. To test out these methods I created several different Integer ArrayLists to pass through these methods and analyze the answers

List of odd numbers:

```
ArrayList<Integer> sampleNumbersOdd = new ArrayList<>();  
sampleNumbersOdd.add(1);  
sampleNumbersOdd.add(3);  
sampleNumbersOdd.add(5);  
sampleNumbersOdd.add(7);  
sampleNumbersOdd.add(9);  
sampleNumbersOdd.add(11);  
sampleNumbersOdd.add(13);
```

List of Even Numbers:

```
ArrayList<Integer> sampleNumbersEven = new ArrayList<>();  
sampleNumbersEven.add(2);  
sampleNumbersEven.add(4);  
sampleNumbersEven.add(6);  
sampleNumbersEven.add(8);  
sampleNumbersEven.add(10);  
sampleNumbersEven.add(12);
```

List of numbers for Mode method:

```
ArrayList<Integer> sampleNumbersMode = new ArrayList<>();  
sampleNumbersMode.add(1);  
sampleNumbersMode.add(3);  
sampleNumbersMode.add(5);  
sampleNumbersMode.add(8);  
sampleNumbersMode.add(4);  
sampleNumbersMode.add(3);  
sampleNumbersMode.add(6);  
sampleNumbersMode.add(7);  
sampleNumbersMode.add(1);  
sampleNumbersMode.add(9);  
sampleNumbersMode.add(8);  
sampleNumbersMode.add(3);
```

For the mean, median, mode, and standard deviation method this is the output:

```
The Mean is 7.0  
The Median with even input is 7.0  
The Median with odd input is 7.0  
The Mode is 3  
The standard deviation of the Population is 3.415650255319866
```

For the mean method, the odd sample numbers were used. Both samples were used for the two median outputs. For the mode, the sample numbers meant for mode were used. For standard deviation, the even sample numbers were used.

The factorial method was called for the number 9:

```
factorial( n: 9));
```

And this is the output:

```
the factorial is: 362880
```

The permutation and combination methods were called with the same inputs:

```
permutations( n: 10, k: 3));  
combinations( n: 10, k: 3));
```

And these were the outputs:

```
The number of permutations is 720  
The number of combinations is 120
```

These were the probabilities passed through the isIndependentt and isDependent methods:

```
+ isIndependent( pA: 0.5, pB: 0.5, pAGivenB: 0.5, pBGivenA: 0.5, pAAndB: 0.25));  
isDependent( pA: 0.5, pB: 0.5, pAGivenB: 0.5, pBGivenA: 0.5, pAAndB: 0.25));;
```

And these were the outputs:

```
Is the problem independent? true  
Is the problem dependent? false
```

For conditional probability the numerator and denominator values were passed in:

```
conditionalProbability( pAAndB: .2, pB: .4));
```

And this is the output:

```
he conditional probability is 0.5
```

The values passed into the binomial probability distribution method:

```
binomialProbabilityDistribution(p: 0.8, q: 0.2, n: 10, y: 7);
```

The output:

```
The binomial probability distribution is 0.2013265920000001
```

The values passed into the geometric probability distribution method:

```
geometricProbabilityDistribution(q: 0.8, p: 0.2, y: 5);
```

The output:

```
The geometric probability distribution is 0.08192000000000002
```